The Evolution of Deep Knowledge Tracing: From Markov Chains to Cognitive Transformers

1. Classical Foundations (1960 – 1994)

- Rasch 1PL / 2PL IRT
  [1960 / 1980]
  Psychometric logistic curves
  $P(\text{correct}) = \sigma(\theta_s - b_q)$
- Bayesian Knowledge Tracing (BKT)
  [Corbett & Anderson '94]
  Hidden Markov Model per skill
  Binary mastery state $\{0, 1\}$

2. Recurrent Deep KT (2015 – 2018)

- Deep Knowledge Tracing (DKT)
  [Piech et al., NeurIPS '15]
  First deep model; LSTM/RNN
  Compresses history into h_t
- DKT+ Regularization
  [Yeung & Yeung, L@S '18]
  Fixes non-monotonic waviness
  and reconstruction errors

3. Memory-Augmented (2017 – 2019)

- DKVMN
  [Zhang et al., WWW '17]
  Static Key Matrix M^k
  Dynamic Mastery Matrix M_t^v
- SKVMN
  [Abdelrahman & Wang '19]
  Sequential key-value hops
  Triangle LSTM integration

4. Self-Attentive / Transf. (2019 – 2023)

- SAKT
  [Pandey & Karypis, EDM '19]
  Self-attention over past answers
  Fast parallelized matrix training
- SAINT / SAINT+
  [Choi et al., EDM '20]
  Separated encoder (exercises)
  and decoder (responses)
- simpleKT
  [Liu et al., ICLR '23]
  Dot-product query-key matching
  Eliminates feed-forward stack

5. Context & Cognition (2020 – 2024)

- AKT (Context-Aware KT)
  [Ghosh et al., KDD '20]
  1PL Rasch difficulty embeddings
  Monotonic Ebbinghaus decay kernel
- CKT
  [Shen et al., SIGIR '20]
  1D causal CNN feature extraction
  Individualized learning rates
- SynapseKT (Dual-Process)
  [Proposed in This Work]
  Local working memory (CNN) +
  Global long-term retention (Attn)

Key Architectural Trade-Offs at a Glance:

- Recurrence (DKT/DKT+): Hidden state entanglement; fast $\mathcal{O}(1)$ step inference, but opaque mastery and non-monotonic 'waviness'.
- Memory Networks (DKVMN/SKVMN): Explicit concept slots, but suffers severe slot collision when concept count $C \gg N_m$ (e.g. Statics2011).
- Transformers (SAKT/SAINT/simpleKT): High throughput $\mathcal{O}(1)$ training; treats history symmetrically without recency bias; struggles on rare items.
- Context & Cognitive Models (AKT/SynapseKT): Rasch difficulty regularization prevents tail overfitting (97.5% param drop on Algebra05); monotonic decay aligns with forgetting.